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Research on educational big data analysis based on Hive and Spark

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***Abstract*—With the increasing number of students in colleges and universities, a large amount of educational big data has been accumulated in daily teaching work. How to make full use of educational big data to serve education and teaching has theoretical and practical significance. In order to mine the hidden value of educational big data and improve teaching quality, this paper processes and analyzes educational big data based on Hive data warehouse technology. First, the educational big data is collected from the Oracle database to the data warehouse through the Sqoop tool, and the analysis results are obtained through data cleaning, data transformation, data statistics and other steps. Hive and Spark are used in data analysis and statistics. The analysis results are stored to the MySQL and Redis, and finally the analysis results are visualized through the front-end and back-end separation technology.**

Educational big data; Hive; Spark; data mining; data visulization; Sqoop; data warehouse

I. INTRODUCTION

With the rapid development of the times, data is accumulating all the time, and massive data has become an important resource [1]. The deep integration of big data technology and various industries promotes the high-quality development of the industry, including the education industry. Every year, there will be new students entering in the school, and graduates leaving with backpacks. The learning track of students will be recorded in the educational administration system, which accumulates a huge amount of education big data.

The educational administration system stores the relevant data of students' learning, and it is necessary to mine the information for teaching. The post demand of enterprises is changing constantly, and students should be able to better respond to the market demand [2]. Colleges will adjust the corresponding curriculum arrangement in real-time according to the enterprise demand. Facing the new curriculum, students with poor learning ability may not keep up with the teacher's progress. It is particularly important to understand the students' learning situation in real-time through the analysis of educational big data, and adjust the teaching arrangement in real-time. Some students are only limited to their grades or the ranking of their grades in the class, and do not know their level in the college. Through the analysis of learning data to help students' learning status is conducive to students' future career planning. The types of courses in the college are also complex, and students' acceptance of different courses is also very different. The teaching methods of different teachers for the same courses may also be very different. The analysis of educational administration data is helpful to the management of college teaching. For colleges and universities, educational administration is a part of their daily school affairs, and has the characteristics of strong sense of time, heavy tasks, and high accuracy requirements. It is often necessary to take into account educational and teaching resources, student information, student examination information, teaching plans, and other aspects. How to make good use of these data is before every teaching manager.

In order to fully explore the value contained in the educational big data and improve the teaching quality, educational big data analysis and visualization system based on Hive and Spark is designed and implemented. The system is implemented in multi-layers. It migrates data by Sqoop, analyses data by Hive and Spark, shares data by MySQL and Redis, provides data interface by Spring Boot, and visualizes data by Echarts. The actual operation of the system proves that the analysis results are helpful to find problems in teaching and provide suggestions for improving teaching.

II. SYSTEM ARCHITECTURE

The system architecture is shown in Figure 1, consisting of source data layer, data warehouse layer, data share layer, data interface layer and data visualization layer. The source data layer stores educational big data, which is realized by Oracle database, mainly including grades, curriculum, student basic information, lecture, teacher basic information and other information. The source data is extracted into data warehouse through an ETL tool. Hadoop distributed file system (HDFS), which is the most data storage solution for data warehouse, is used to store the massive educational data [3]. In this layer, the offline data analysis and statistics is implemented by Hive and Spark. Hive is a data warehouse tool based on Hadoop, which can extract, transform, store and query massive data [4]. The hive data warehouse tool provides an SQL query interface, which can convert SQL statements into MapReduce tasks for execution. Spark is a fast and universal computing engine designed for large-scale data processing, which can meet the offline and online computing of massive data [5]. The results of analysis and statistics based on Hive and Spark are stored on HDFS, however various applications cannot directly obtain data from HDFS. In order to enable various applications to easily obtain the computation results, the results are stored in the data sharing layer. The data sharing layer is implemented by relational databases MySQL and Redis. Redis is an open source and memory-based Key-Value database. Based on the high-performance caching mechanism, Redis can meet the application scenarios of high concurrency and high throughput

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[6]. The data interface layer is based on Restful and provides query results in JSON format. Restful is a design style and development method of network applications. It is based on HTTP and uses JSON format to define data [7]. The data visualization layer is composed of web pages and WeChat applets. It allows users to understand the phenomena and laws contained in the data through charts and other visualization means.

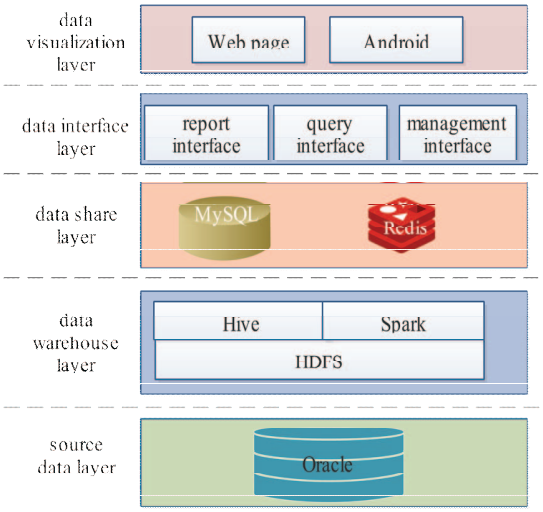

Figure 1. The system architecture

III. DATA ANALYSIS AND VISUALIZATION

As shown in Figure 2, the proposed algorithm is divided into two main steps: data analysis and data visualization.

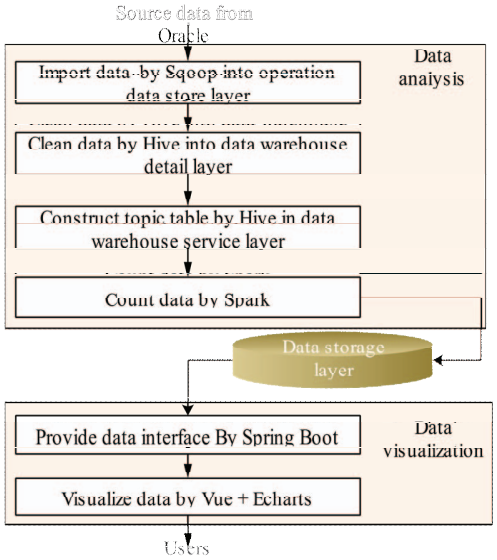

Figure 2. The processing flow

Data analysis is based on Hive and Spark, and the final statistical results are stored in the data sharing layer, namely MySQL and Redis. The data visualization is realized through the front-end and back-end separation technology, and the data interaction between the front-end and back-end is realized through Axios. The back-end realizes data interface through Spring Boot + MyBatis, and the front-end realizes data visualization through Vue + Echarts.

A. Data Analysis based on Hive and Spark

Based on Hive, the data warehouse constructs a three sublayers architecture of operation data store (ODS) layer, data warehouse detail (DWD) layer, and data warehouse service (DWS) layer to analyze educational big data. By data layering, clear data structure can be realized, and anomalies of original data can be shielded, which can simplify complex problems. In the Hive data warehouse, the data of different layers is stored in different databases. External tables are created in the system, and some tables are processed by mirror partition. For example, the score table is divided by semester, and the student basic information table is divided by year of enrollment, improving the query efficiency.

1) ODS layer

The ODS layer stores the educational big data from the source Oracle intact, also known as the data preparation area. These data are the source of subsequent processing data. The Sqoop tool is used for data extraction from Oracle to Hive. Sqoop is an open-source Extract-Transform-Load (ETL) tool that can transfer data between Hive data warehouse and traditional relational data. The underlying layer of Sqoop relies on the MapReduce program to connect to the relational database through JDBC to acquire data in batches. The extraction of educational big data is divided into two steps. The first step is full extraction, which will load all data into the data warehouse. Later, incremental extraction is used to extract the updated data from the educational administration database and update it to the data warehouse. It is implemented by monitoring the database log.

The table name of the ODS layer is named in the form of "O_ source table name". During import, all char and varchar types are converted to string type, number type to double type, date type and TIMESTAMP type remain unchanged.

2) DWD layer

The main functions of DWD layer are: (1) To clean the data of the ODS layer. (2) Perform dimension degradation and dimension modeling on business data tables to reduce the inefficiency caused by multi-table join in subsequent queries.

In the data cleaning stage, the fields with null values, incomplete contents and inconsistent contents are supplemented and corrected. The default value cleaning is completed by removing, filling and other means according to the proportion of missing values. If the learning score is empty, it is supplemented by 0. The format content cleaning includes the date format, the score and other formats. Logic error cleaning is to clear unreasonable values, such as scores that are not numeric or too large.

The basic data is dimensioned according to the star model. Through the four processes of business process selection, granularity declaration, dimension determination and fact

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determination, the learning fact table and teaching fact table are constructed driven by the business process.

3) DWS layer

The DWS summarizes and aggregates the data in the data detail layer according to different subject angles to obtain statistical data. The statistical characteristics are (1) to determine the fields around the needs of educational data analysis. (2) Data is aggregated by different dimensions. (3) The requirements are classified and divided into multiple tables according to the statistical subject.

B.Data Visualization Based On Front-end and Back-end Separation

The back-end provides a data interface, which is implemented through Spring Boot + MyBatis. Spring Boot is a new framework provided by the Pivotal team, which can quickly build and implement Web applications [9]. MyBatis is a persistence layer framework based on Java, which can quickly implement database read and write operations [10]. The implementation of the back-end is mainly completed through the hierarchical design of Controller, Service and Dao. The Controller layer corresponds to the control of business processes. In this layer, the data required by the front-end is obtained by calling the interface of the Service layer, and the data is converted into JSON format data and returned to the front-end. The service layer corresponds to the business application logic design. Encapsulating the service layer business logic is conducive to reducing the duplication of programs and improving maintainability. The Dao layer is responsible for the work of the data persistence layer, and the data accessing from MySQL and Redis is implemented by MyBatis. The back-end implementation architecture is shown in Figure 3.

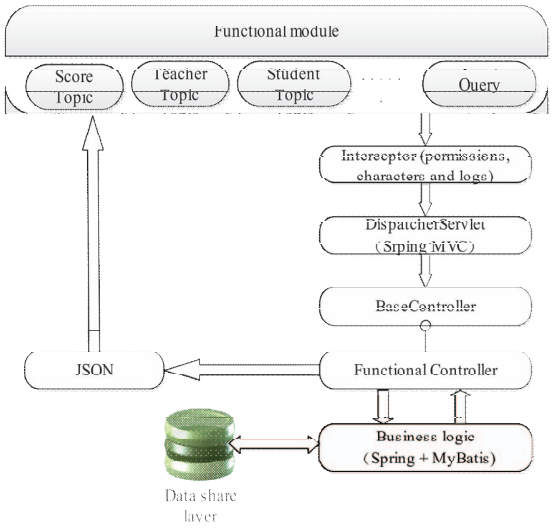

Figure 3. The back-end architecture based on Spring Boot + MyBatis

The front-end and back-end transfer data in JSON format, and the front-end calls the back-end data through Axios which is an Ajax interface. The realization of visualization is mainly based on Echarts technologies. Echarts is a JavaScript based data visualization chart library that provides intuitive and customizable data visualization charts [11].

IV.SYSTEM OPERATION ANALYSIS

The system has been running and applied to teaching decision-making. The complete system is deployed on five servers. Three servers deploy Hadoop, Hive and Spark to form a data warehouse cluster. MySQL, Redis and back-end application are installed on one server, and front-end application is deployed on the other server.

The system is mainly composed of three topics: scores, students and teachers. Each topics analyzes the data from multiple dimensions. Only a small part of the visualization results is displayed in this paper.

A.Scores Topic

The scores topic is based on the students' learning achievements, and analyzes the students' learning from different dimensions such as the differences of students' learning, the relevance of courses, and the differences of teachers' assessment of courses. Figure 4 shows the heatmap of the course correlation of software technology in 2018, which is calculated from the scores of each course of the major students.

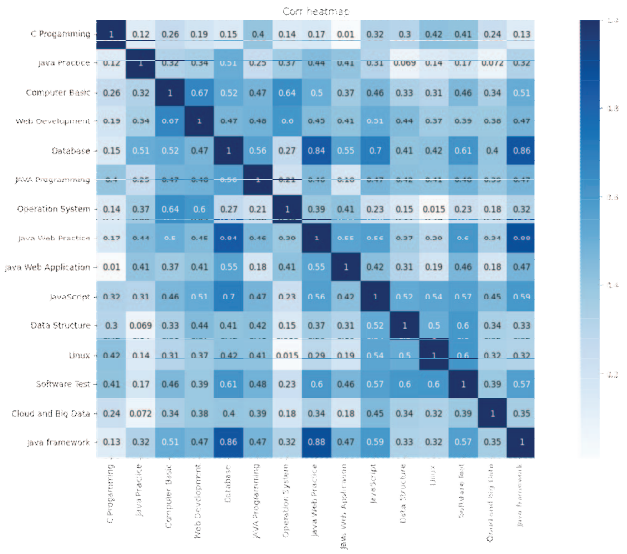

Figure 4. The course correlation heatmap of the 18th software technology

Step 1: Import the score data. It imports the data from the Oracle database to Education_ODS database in Hive data warehouse by Sqoop.

Step 2: Clean the data through Hive. During the cleaning process, different scores are uniformly converted into hundred-point format, and the cleaned data is stored in Education_ DWD database.

Step 3: Build a topic table of score. It forms a wide table by associating multiple tables such as score table, curriculum table, student table and teacher table.

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Step 4: Analyze the score topic table based on Spark. First, it extracts the scores of different years and majors into sub-tables. Then, it converts the course name into a column, and the column value is the corresponding score. Lastly, it performs correlation analysis on the columns of the table, and stores the analysis results in the MySQL database.

Step 5: Provide a course relevance query interface based on Spring Boot. The input parameters of the interface are the major name and year of enrollment, and the output parameters are the course correlation matrix of the major.

Step 6: Display the course correlation heatmap based on Echarts.

Figure 5 shows the heatmap of the course correlation of software technology in 2019. By comparing Figure 4, it can be seen that the relevance between the courses of software technology in 2019 is stronger than that in 2018. Because the talent training program and curriculum standards of the software technology in 2019 have been greatly reformed and standardized. It can better reflect the consistency and integrity of professional talent training.

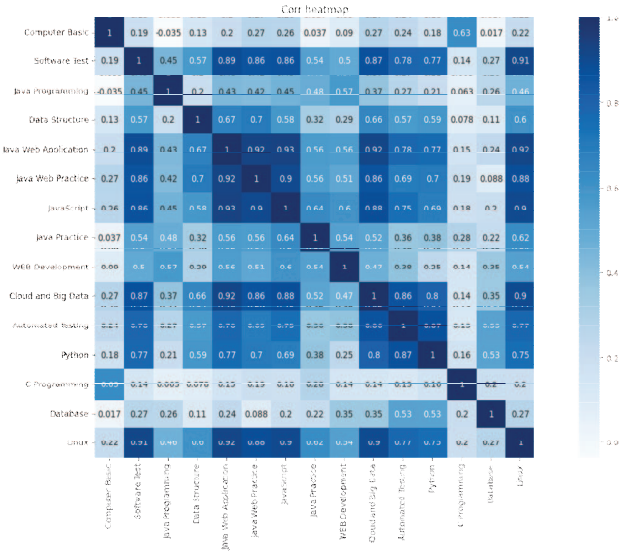

Figure 5. The course correlation heatmap of the 19th software technology

Figure 6 shows the number of courses with different examination failure rates. The examination failure rate of 252 courses is close to zero, and the number of courses with a failure rate of more than 1/3 is small. Figure 6 shows that the difficulty of curriculum assessment in the institute is not hard, which is not conducive to the realization of higher vocational education objectives. In order to make more students take their studies seriously, it is necessary for the college to implement the principle of lenient entry and strict exit. Through strict requirements, students can improve their skills in a down-to-earth manner, not only can they graduate successfully, but also can take the road of skill development and ingenuity to build dreams.

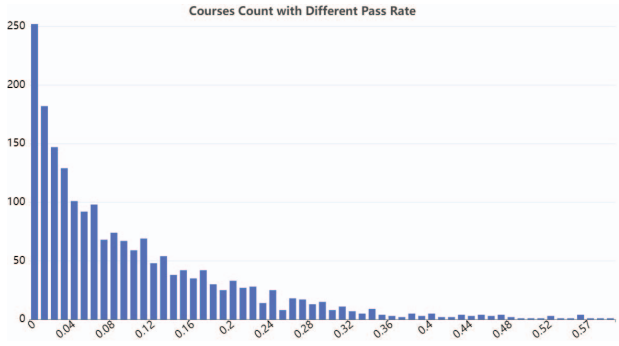

Figure 6. The number of courses with different examination failure rates

B.Students Topic

The student topic takes the student information as the source, and analyzes the student data from multiple dimensions such as student origin, student gender, different years, and enrollment type. Figure 7 shows the enrollment situation in different years.

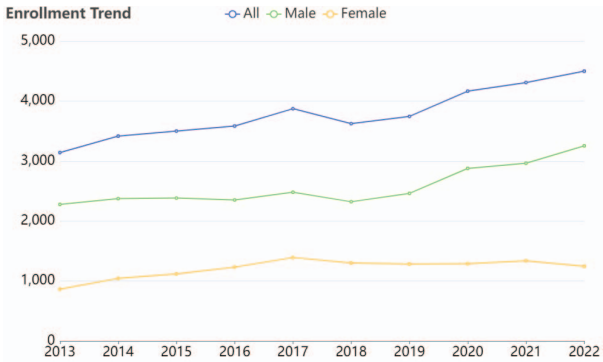

Figure 7. The enrollment situation in different years

Since 2013, the enrollment of the institute has increased year by year, and the proportion of male and female students has changed with the enrollment of mainstream majors. The navigation technology and marine engineering technology are backbone majors, and the proportion of male students has increased with good enrollment. Figure 8 shows the number of enrolled students in Jiangsu province and non-Jiangsu provinces in different years. It can be seen that the proportion of non-Jiangsu students is increasing year by year.

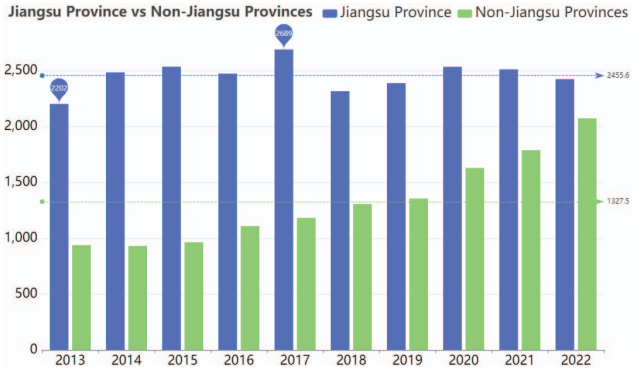

Figure 8. The enrolled students number in different areas and years

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Figure 9 shows the number of students enrolled by different type in 2022. Through the analysis of student topic data, it can be found that the student origin is various. The teaching ecosystem of the institute must undergo purposeful and reasonable changes to effectively meet the challenges.

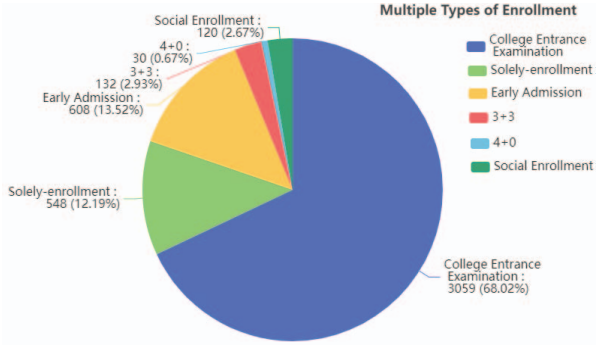

Figure 9. The number of students enrolled by different type in 2022

C.Teachers Topic

The teacher topic is based on teacher information and analyzed from multiple dimensions such as teacher type, professional ranks and teacher gender. Figure 10 shows the number of full-time and part-time teachers in different colleges of the institute. It can be seen the number of part-time teachers in most colleges is greater than that of full-time teachers, and the majority of part-time teachers come from enterprises, which also shows that the institute pays attention to the integration of production and teaching.

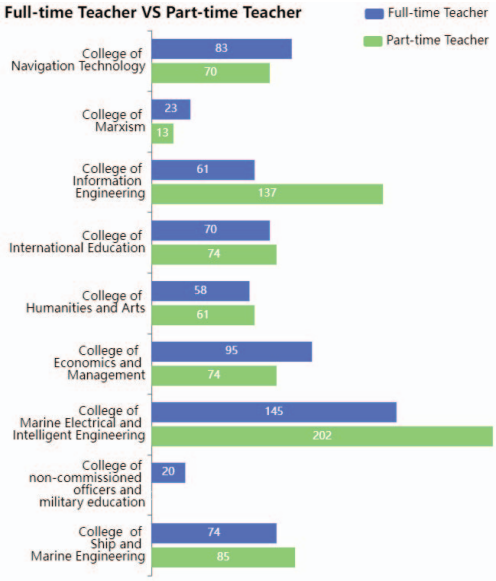

Figure 10. The number of students enrolled by different type in 2022

V.CONCLUSION

The rapid development of big data technology has made it applied to various fields, and universities and colleges have also made full use of big data technology to improve the intelligent level of campus. The educational big data accumulated over time and has become a valuable data asset. This paper is based on Hive and Spark technology to mine education big data through data extraction, cleaning, analysis, statistics and visualization steps. The actual operation of the system has proved that it is helpful to improve the intelligent management level and teaching quality.

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